

جامعة أم القرى

UMM AL-QURA UNIVERSITY

# Lost and Found Management System

Database Systems Project

| | |
|---|---|
| **Division Number:** | 7 ( Practical ) |
| **Course Name & Code:** | Introduction to databases ( DS2201 ) |
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# Task distribution table

| Student | Contribution Description |
|---|---|
| 1. Layan Mohammed Alshahrani | • Writing the Scenario (Introduction, Entities, Attributes, Relationships description).<br>• Create the database (LostAndFound).<br>• Create the User table with appropriate data types.<br>• Write and test TWO basic SELECT queries.<br>Include column selection and WHERE conditions. |
| 2. Ethar Hatem Tamim | • Designing and Drawing the ER Diagram.<br>• Create the Report table.<br>• Establish the relationship (Foreign Key) between Report and User tables.<br>• Write and test TWO aggregate function queries Use functions like COUNT.<br>Include GROUP BY. |
| 3. Azizah Al-Ajyan | • Converting ERD into Relational Schema.<br>• Add constraints (Primary Key, NOT NULL, CHECK).<br>• Insert 5 records into each table (INSERT statements).<br>• Write and test ONE Subquery<br>Answer a specific question using a nested query<br>• Create ONE View<br>Generate a simple report based on the database |
| 4. Ruwaina Majid Alharbi | • Formatting the report (University template, Table of Contents, Page numbering).<br>• Create the Item table Link Item table with Report table.<br>• Add UNIQUE constraint on Report_ID.<br>• Write and test ONE JOIN query Combine data from at least two tables. |

# Scenario

## 1. System Description

The Lost and Found Management System is designed to help users report lost items and manage found items efficiently. The system allows users to submit reports about lost items, where each report contains detailed information about the item. The main objective of the system is to organize and track lost items in a structured database.

## 2. Entities

### 2.1 User

**Attributes:**

User_ID (Primary Key)
Fname
Lname
Email
Phone
Role

### 2.2 Report

**Attributes:**

Report_ID (Primary Key)
Report_Date
Report_Type
Report_Status
Pickup_Location

### 2.3 Item

**Attributes:**

Item_ID (Primary Key)
Item_Name
Category
Description
Status

## 3. Relationships

### 3.1 Submits (User – Report)

One User can submit many Reports (1:M). Each Report is submitted by exactly one User.

### 3.2 Includes (Report – Item)

Each Report includes one Item, and each Item belongs to one Report (1:1).

# ER Diagram

An ER Diagram was created using a diagram tool (draw.io).
The diagram includes three entities (User, Report, Item) and two relationships (Submits and Includes).
Primary keys are underlined in the diagram, and the Name attribute in User is represented as a composite attribute (Fname and Lname).

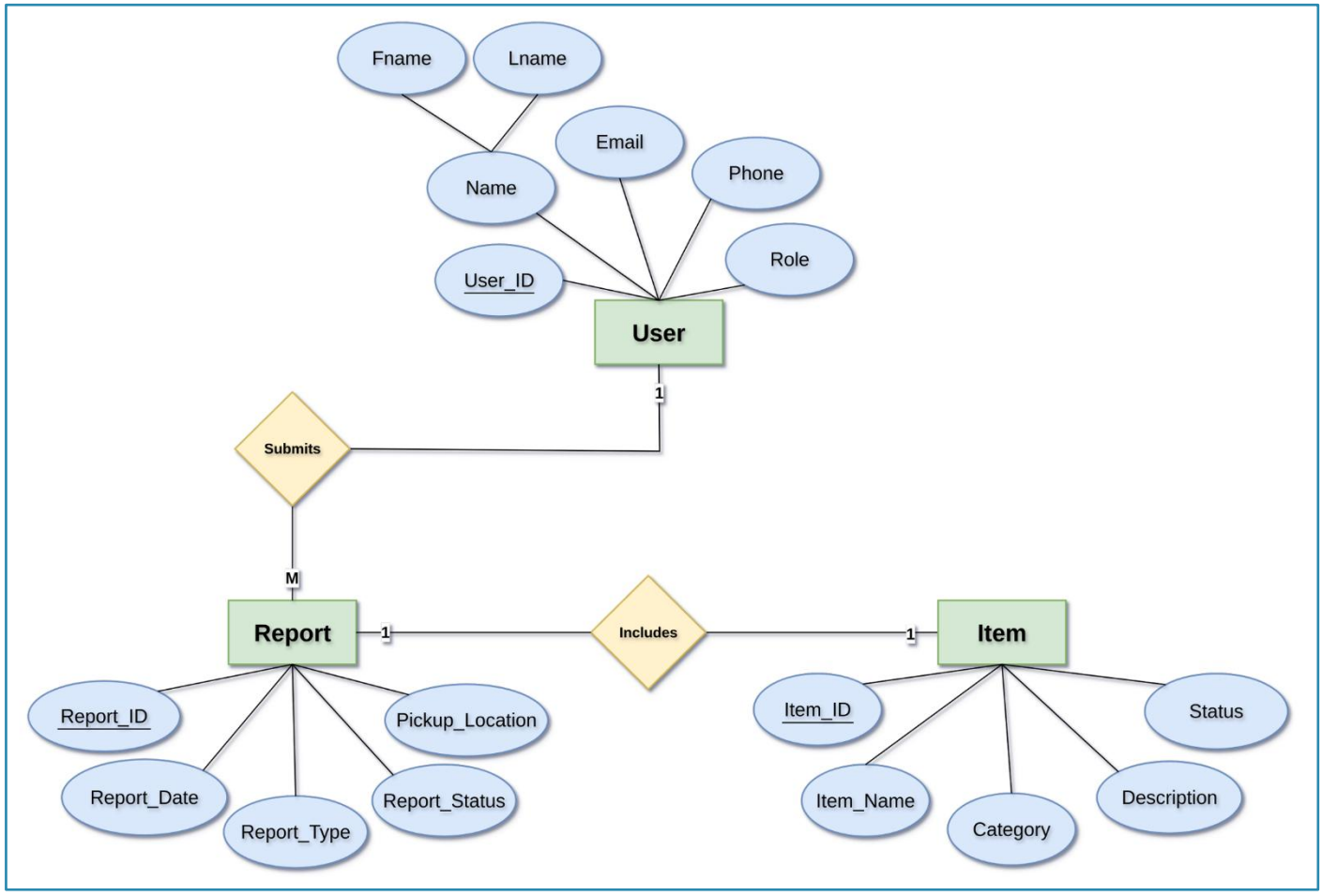



*Figure 1*

# Relational Schema

User (User_ID PK, Fname, Lname, Email, Phone, Role)

Report (Report_ID PK, Report_Date, Report_Type, Report_Status, Pickup_Location, User_ID FK)

Item (Item_ID PK, Item_Name, Category, Description, Status, Report_ID FK UNIQUE)

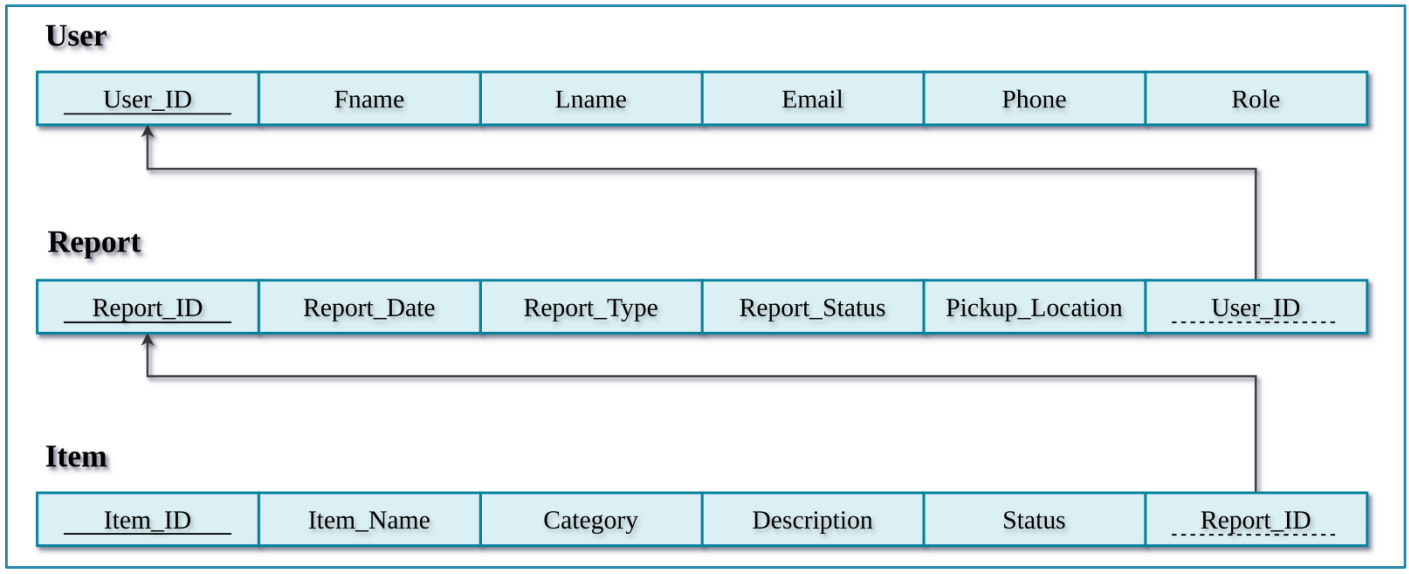



*Figure **2***

# Database Creation

```
1   CREATE DATABASE LostAndFound;
2   SHOW DATABASES;
3   USE LostAndFound;
4
5
```

Result Grid | Filter Rows:

| Database |
|---|
| ethar |
| information_schema |
| lostandfound |
| mysql |
| najddelights |
| performance_schema |
| sys |

*Figure 3*

# Tables Creation

```
5   CREATE TABLE User (
6   User_ID INT,
7   Fname VARCHAR(60),
8   Lname VARCHAR(60),
9   Email VARCHAR(100),
10  Phone VARCHAR(20),
11  Role ENUM('User','Admin')
12  );
13  DESCRIBE User;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| User_ID | int | YES | | NULL | |
| Fname | varchar(60) | YES | | NULL | |
| Lname | varchar(60) | YES | | NULL | |
| Email | varchar(100) | YES | | NULL | |
| Phone | varchar(20) | YES | | NULL | |
| Role | enum('User','Admin') | YES | | NULL | |

*Figure 4*

```
15  CREATE TABLE Report (
16  Report_ID INT,
17  Report_Date DATE,
18  Report_Type VARCHAR(20),
19  Report_Status ENUM('Pending', 'Resolved', 'Cancelled'),
20  Pickup_Location VARCHAR(100),
21  User_ID INT
22  );
23  DESCRIBE Report;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| Report_ID | int | YES | | NULL | |
| Report_Date | date | YES | | NULL | |
| Report_Type | varchar(20) | YES | | NULL | |
| Report_Status | enum('Pending','Resolved','Cancelled') | YES | | NULL | |
| Pickup_Location | varchar(100) | YES | | NULL | |
| User_ID | int | YES | | NULL | |

*Figure 5*

```
25  CREATE TABLE Item (
26  Item_ID INT,
27  Item_Name VARCHAR(100),
28  Category VARCHAR(50),
29  Description VARCHAR(100),
30  Status VARCHAR(20),
31  Report_ID INT
32  );
33  DESCRIBE Item;
```

Result Grid | Filter Rows: | Export: | Wra

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| Item_ID | int | YES | | NULL | |
| Item_Name | varchar(100) | YES | | NULL | |
| Category | varchar(50) | YES | | NULL | |
| Description | varchar(100) | YES | | NULL | |
| Status | varchar(20) | YES | | NULL | |
| Report_ID | int | YES | | NULL | |

*Figure 6*

# Constraints and Relationships

```
ALTER TABLE User
ADD PRIMARY KEY (User_ID);

ALTER TABLE Report
ADD PRIMARY KEY (Report_ID);

ALTER TABLE Item
ADD PRIMARY KEY (Item_ID);

ALTER TABLE Report
ADD FOREIGN KEY (User_ID) REFERENCES User(User_ID);

ALTER TABLE Item
ADD UNIQUE (Report_ID);

ALTER TABLE Item

ADD FOREIGN KEY (Report_ID) REFERENCES Report(Report_ID);

SHOW CREATE TABLE User;
SHOW CREATE TABLE Report;
SHOW CREATE TABLE Item;

ALTER TABLE User
ADD UNIQUE (Email);

ALTER TABLE User
MODIFY COLUMN Fname VARCHAR(60) NOT NULL,
MODIFY COLUMN Lname VARCHAR(60) NOT NULL;

ALTER TABLE Report
MODIFY COLUMN Report_Date DATE NOT NULL,
MODIFY COLUMN Report_Status ENUM('Pending','Resolved','Cancelled') NOT NULL,
MODIFY COLUMN User_ID INT NOT NULL;
ALTER TABLE Item
MODIFY COLUMN Item_Name VARCHAR(100) NOT NULL,
MODIFY COLUMN Status ENUM('Lost','Found') NOT NULL,
MODIFY COLUMN Report_ID INT NOT NULL;

ALTER TABLE Item
ADD CHECK (Status IN ('Lost','Found'));

ALTER TABLE Report
ADD CHECK (Report_Status IN ('Pending','Resolved','Cancelled'));
```

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| User_ID | int | NO | PRI | NULL | |
| Fname | varchar(60) | NO | | NULL | |
| Lname | varchar(60) | NO | | NULL | |
| Email | varchar(100) | YES | UNI | NULL | |
| Phone | varchar(20) | YES | | NULL | |
| Role | enum('User','Admin') | YES | | NULL | |

*Figure 7*

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| Report_ID | int | NO | PRI | NULL | |
| Report_Date | date | NO | | NULL | |
| Report_Type | varchar(20) | YES | | NULL | |
| Report_Status | enum('Pending','Resolved','Cancelled') | NO | | NULL | |
| Pickup_Location | varchar(100) | YES | | NULL | |
| User_ID | int | NO | MUL | NULL | |

*Figure 8*

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| Item_ID | int | NO | PRI | NULL | |
| Item_Name | varchar(100) | NO | | NULL | |
| Category | varchar(50) | YES | | NULL | |
| Description | varchar(100) | YES | | NULL | |
| Status | enum('Lost','Found') | NO | | NULL | |
| Report_ID | int | NO | UNI | NULL | |

*Figure 9*

# Data Insertion

```
83 •  INSERT INTO User VALUES
84    (1,'Layan','Mohammed','layan@email.com','0501111111','Admin'),
85    (2,'Ethar','Hatem','ethar@email.com','0502222222','User'),
86    (3,'Azizah','Al-Ajyan','azizah@email.com','0503333333','User'),
87    (4,'Ruwaina','Majid','ruwaina@email.com','0504444444','User'),
88    (5,'Noor','Ali','noor@email.com','0505555555','User');
```

Result Grid | Filter Rows: | Edit: | Export/Import:

| User_ID | Fname | Lname | Email | Phone | Role |
|---|---|---|---|---|---|
| 1 | Layan | Mohammed | layan@email.com | 0501111111 | Admin |
| 2 | Ethar | Hatem | ethar@email.com | 0502222222 | User |
| 3 | Azizah | Al-Ajyan | azizah@email.com | 0503333333 | User |
| 4 | Ruwaina | Majid | ruwaina@email.com | 0504444444 | User |
| 5 | Noor | Ali | noor@email.com | 0505555555 | User |
| NULL | NULL | NULL | NULL | NULL | NULL |

*Figure 10*

```
90 •  INSERT INTO Report VALUES
91    (2,'2025-01-02','Found','Resolved','Gate B',2),
92    (3,'2025-01-03','Lost','Pending','Library',3),
93    (4,'2025-01-04','Found','Resolved','Cafeteria',4),
94    (5,'2025-01-05','Lost','Pending','Parking',5);
```

Result Grid | Filter Rows: | Edit: | Export/Import:

| Report_ID | Report_Date | Report_Type | Report_Status | Pickup_Location | User_ID |
|---|---|---|---|---|---|
| 2 | 2025-01-02 | Found | Resolved | Gate B | 2 |
| 3 | 2025-01-03 | Lost | Pending | Library | 3 |
| 4 | 2025-01-04 | Found | Resolved | Cafeteria | 4 |
| 5 | 2025-01-05 | Lost | Pending | Parking | 5 |
| NULL | NULL | NULL | NULL | NULL | NULL |

*Figure 11*

```
96 •  INSERT INTO Item VALUES
97    (2,'Wallet','Personal','Black wallet','Found',2),
98    (3,'Bag','Accessories','Backpack','Lost',3),
99    (4,'Keys','Personal','Car keys','Found',4),
100   (5,'Watch','Accessories','Silver watch','Lost',5);
```

Result Grid | Filter Rows: | Edit: | Exp

| Item_ID | Item_Name | Category | Description | Status | Report_ID |
|---|---|---|---|---|---|
| 2 | Wallet | Personal | Black wallet | Found | 2 |
| 3 | Bag | Accessories | Backpack | Lost | 3 |
| 4 | Keys | Personal | Car keys | Found | 4 |
| 5 | Watch | Accessories | Silver watch | Lost | 5 |
| NULL | NULL | NULL | NULL | NULL | NULL |

*Figure12*

# Data Retrieval

Shows SELECT queries used to display all records from the User, Report, and Item tables.

Output

Action Output

| # | Time | Action | Message | Duration / Fetch |
|---|---|---|---|---|
| 1 | 20:24:04 | CREATE DATABASE LostAndFound | 1 row(s) affected | 0.000 sec |
| 2 | 20:24:04 | SHOW DATABASES | 7 row(s) returned | 0.000 sec / 0.000 sec |
| 3 | 20:24:04 | USE LostAndFound | 0 row(s) affected | 0.000 sec |
| 4 | 20:24:04 | CREATE TABLE User ( User_ID INT, Fname VARCHAR(60), Lname VARCHAR(6... | 0 row(s) affected | 0.016 sec |
| 5 | 20:24:04 | DESCRIBE User | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 6 | 20:24:04 | CREATE TABLE Report ( Report_ID INT, Report_Date DATE, Report_Type VAR... | 0 row(s) affected | 0.016 sec |
| 7 | 20:24:04 | DESCRIBE Report | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 8 | 20:24:04 | CREATE TABLE Item ( Item_ID INT, Item_Name VARCHAR(100), Category VARC... | 0 row(s) affected | 0.000 sec |
| 9 | 20:24:04 | DESCRIBE Item | 6 row(s) returned | 0.000 sec / 0.000 sec |
| 10 | 20:24:04 | SHOW TABLES | 3 row(s) returned | 0.000 sec / 0.000 sec |
| 11 | 20:24:04 | ALTER TABLE User ADD PRIMARY KEY (User_ID) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.031 sec |
| 12 | 20:24:04 | ALTER TABLE Report ADD PRIMARY KEY (Report_ID) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.032 sec |
| 13 | 20:24:04 | ALTER TABLE Item ADD PRIMARY KEY (Item_ID) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.032 sec |
| 14 | 20:24:04 | ALTER TABLE Report ADD FOREIGN KEY (User_ID) REFERENCES User(User_I... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.031 sec |
| 15 | 20:24:04 | ALTER TABLE Item ADD UNIQUE (Report_ID) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.031 sec |
| 16 | 20:24:04 | ALTER TABLE Item ADD FOREIGN KEY (Report_ID) REFERENCES Report(Rep... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.015 sec |

*Figure 13*

Output

Action Output

| # | Time | Action | Message | Duration / Fetch |
|---|---|---|---|---|
| 17 | 20:24:04 | SHOW CREATE TABLE User | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 18 | 20:24:04 | SHOW CREATE TABLE Report | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 19 | 20:24:04 | SHOW CREATE TABLE Item | 1 row(s) returned | 0.000 sec / 0.000 sec |
| 20 | 20:24:04 | ALTER TABLE User ADD UNIQUE (Email) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.047 sec |
| 21 | 20:24:04 | ALTER TABLE User MODIFY COLUMN Fname VARCHAR(60) NOT NULL, MODI... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.047 sec |
| 22 | 20:24:04 | ALTER TABLE Report MODIFY COLUMN Report_Date DATE NOT NULL, MODI... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.031 sec |
| 23 | 20:24:04 | ALTER TABLE Item MODIFY COLUMN Item_Name VARCHAR(100) NOT NULL, ... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.047 sec |
| 24 | 20:24:04 | ALTER TABLE Item ADD CHECK (Status IN ('Lost','Found')) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.047 sec |
| 25 | 20:24:04 | ALTER TABLE Report ADD CHECK (Report_Status IN ('Pending','Resolved','Canc... | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 | 0.063 sec |
| 26 | 20:24:04 | INSERT INTO User VALUES (1,'Layan','Mohammed','layan@email.com','0501111... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.000 sec |
| 27 | 20:24:04 | INSERT INTO Report VALUES (1,'2025-01-01','Lost','Pending','Gate A',1), (2,'2025... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.000 sec |
| 28 | 20:24:04 | INSERT INTO Item VALUES (1,'Phone','Electronics','iPhone','Lost',1), (2,'Wallet','P... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.000 sec |
| 29 | 20:24:04 | SELECT * FROM User LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec |
| 30 | 20:24:04 | SELECT * FROM Report LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec |
| 31 | 20:24:05 | SELECT * FROM Item LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec |

*Figure 14*

# Basic SELECT

```
102 •   SELECT User_ID , Fname , Lname , Role From User
103     WHERE Role IN ('User');
```

Result Grid | Filter Rows: | Edit:

| User_ID | Fname | Lname | Role |
|---|---|---|---|
| 2 | Ethar | Hatem | User |
| 3 | Azizah | Al-Ajyan | User |
| 4 | Ruwaina | Majid | User |
| 5 | Noor | Ali | User |
| NULL | NULL | NULL | NULL |

*Figure 15*

```
104 •   SELECT Item_ID, Item_Name, Category FROM Item
105     WHERE Status IN ('Lost');
```

Result Grid | Filter Rows: | Edit:

| Item_ID | Item_Name | Category |
|---|---|---|
| 3 | Bag | Accessories |
| 5 | Watch | Accessories |
| NULL | NULL | NULL |

*Figure 16*

# Aggregate Functions

```
107 •   SELECT COUNT(*) AS Total_Reports
108     FROM Report;
```

Result Grid | Filter Rows:

| Total_Reports |
|---|
| 4 |

*Figure 17*

```
110 •   SELECT Status, COUNT(*) AS Total_Items
111     FROM Item
112     GROUP BY Status;
```

Result Grid | Filter Rows: | Export:

| Status | Total_Items |
|---|---|
| Found | 2 |
| Lost | 2 |

*Figure 18*

# Join

```
114 •   SELECT User.Fname, User.Lname, Report.Report_Status
115     FROM User
116     JOIN Report ON User.User_ID = Report.User_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

| Fname | Lname | Report_Status |
|---|---|---|
| Ethar | Hatem | Resolved |
| Azizah | Al-Ajyan | Pending |
| Ruwaina | Majid | Resolved |
| Noor | Ali | Pending |

*Figure 19*

# Subquery

```
118 •   SELECT Item_Name
119     FROM Item
120     WHERE Report_ID IN (SELECT Report_ID FROM Report
121     WHERE Pickup_Location = 'Library');
```

Result Grid | Filter Rows: | Export: | Wrap Cell

| Item_Name |
|---|
| Bag |

*Figure 20*

# View

```
123 •   CREATE VIEW Pending_Reports AS
124     SELECT Report_ID, Pickup_Location, Report_Status
125     FROM Report
126     WHERE Report_Status = 'Pending';
127
128 •   SELECT * FROM Pending_Reports;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

| Report_ID | Pickup_Location | Report_Status |
|---|---|---|
| 3 | Library | Pending |
| 5 | Parking | Pending |

*Figure 21*








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